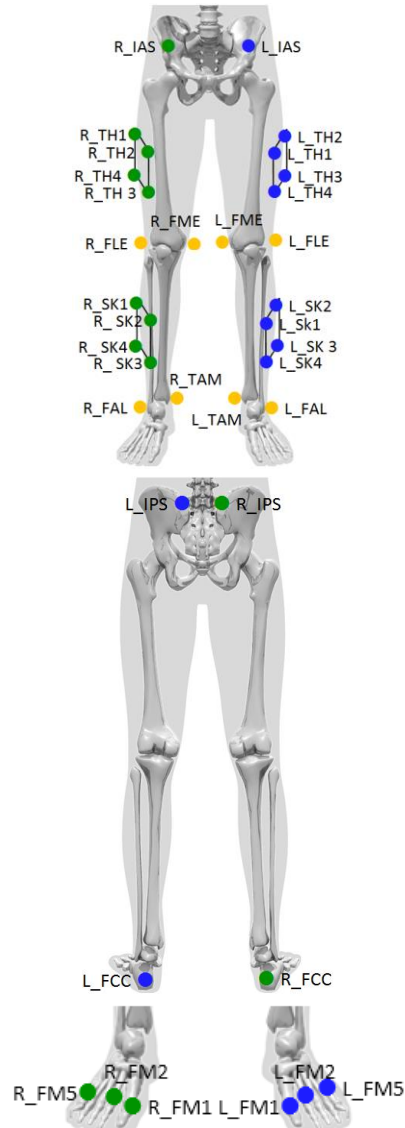


# Qualisys PAF package: CAST lower body marker set<sup>1</sup>

|  |  |                   |                                                          |             |           |
|--|------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------|-------------|-----------|
|  | Name                                                                               | Ref. <sup>2</sup> | Location                                                 | Static (36) | Dyn. (28) |
|  | L_IAS                                                                              | IAS               | Anterior superior iliac spine                            | X           | X         |
|  | L_IPS                                                                              | IPS               | Posterior superior iliac spine                           | X           | X         |
|  | R_IPS                                                                              | IPS               | Posterior superior iliac spine                           | X           | X         |
|  | R_IAS                                                                              | IAS               | Right anterior superior iliac spine                      | X           | X         |
|  | L_TH1-4                                                                            |                   | Cluster                                                  | X           | X         |
|  | L_FLE                                                                              | FLE               | Lateral epicondyle                                       | X           |           |
|  | L_FME                                                                              | FME               | Medial epicondyle                                        | X           |           |
|  | L_SK1-4                                                                            |                   | Cluster                                                  | X           | X         |
|  | L_FAL                                                                              | FAL               | Lateral prominence of the lateral malleolus              | X           |           |
|  | L_TAM                                                                              | TAM               | Medial prominence of the medial malleolus                | X           |           |
|  | L_FCC                                                                              | FCC               | Aspect of the Achilles tendon insertion on the calcaneus | X           | X         |
|  | L_FM1                                                                              | FM1               | Dorsal margin of the first metatarsal head               | X           | X         |
|  | L_FM2                                                                              | FM2               | Dorsal aspect of the second metatarsal head              | X           | X         |
|  | L_FM5                                                                              | FM5               | Dorsal margin of the fifth metatarsal head               | X           | X         |
|  | R_TH1-4                                                                            |                   | Cluster                                                  | X           | X         |
|  | R_FLE                                                                              | FLE               | Lateral epicondyle                                       | X           |           |
|  | R_FME                                                                              | FME               | Medial epicondyle                                        | X           |           |
|  | R_SK1-4                                                                            |                   | Cluster                                                  | X           | X         |
|  | R_FAL                                                                              | FAL               | Lateral prominence of the lateral malleolus              | X           |           |
|  | R_TAM                                                                              | TAM               | Medial prominence of the medial malleolus                | X           |           |
|  | R_FCC                                                                              | FCC               | Aspect of the Achilles tendon insertion on the calcaneus | X           | X         |
|  | R_FM1                                                                              | FM1               | Dorsal margin of the first metatarsal head               | X           | X         |
|  | R_FM2                                                                              | FM2               | Dorsal aspect of the second metatarsal head              | X           | X         |
|  | R_FM5                                                                              | FM5               | Dorsal margin of the fifth metatarsal head               | X           | X         |

<sup>1</sup> Cappozzo, A., Catani, F., Della Croce, U., Leardini, A. (1995). Position and orientation in space of bones during movement: anatomical frame definition and determination. Clinical Biomechanics, 10, 171-178.

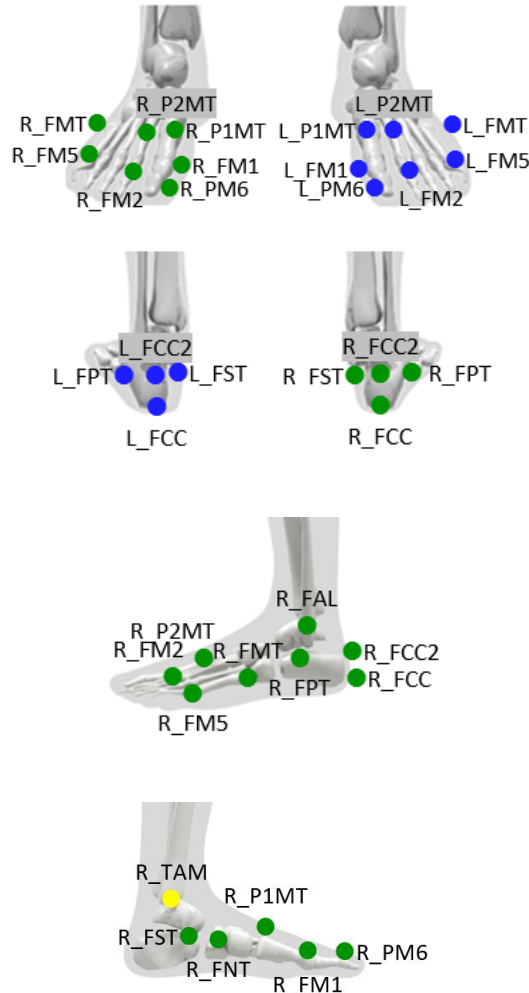
<sup>2</sup> Sint Jan, S. Van (2007). Color Atlas of Skeletal Landmark Definitions. Guidelines for Reproducible Manual and Virtual Palpations. Edinburgh: Churchill Livingstone.

<sup>1</sup> Cappozzo, A., Catani, F., Della Croce, U., Leardini, A. (1995). Position and orientation in space of bones during movement: anatomical frame definition and determination. *Clinical Biomechanics*, 10, 171-178.

<sup>2</sup> Sint Jan, S. Van (2007). *Color Atlas of Skeletal Landmark Definitions. Guidelines for Reproducible Manual and Virtual Palpations*. Edinburgh: Churchill Livingstone.

<sup>3</sup> Stebbins J. Harrington M. Thompson N. Zavatsky A. Theologis T. (2006) Repeatability of a model for measuring multi-segment foot kinematics in children. *Gait Posture*. 2006 Jun;23(4):401-10.

# Qualisys PAF package: CAST<sup>1</sup> lower body marker set (including Rizzoli foot model <sup>3</sup>)

|                                                                                    | Name   | Ref. <sup>2</sup> | Location                                                                                   | Static (24) | Dyn. (24) |
|------------------------------------------------------------------------------------|--------|-------------------|--------------------------------------------------------------------------------------------|-------------|-----------|
|  | L_FCC2 |                   | Proximal end of posterior aspect of calcaneus                                              | X           | X         |
|                                                                                    | L_FCC  | FCC               | Most distal point of the attachment area of the Achilles tendon on the calcaneus           | X           | X         |
|                                                                                    | L_FPT  | FPT               | Lateral apex of the peroneal tubercle                                                      | X           | X         |
|                                                                                    | L_FMT  | FMT               | Base of the fifth metatarsal, dorso-lateral aspect of the fifth metatarso-cuboid joint     | X           | X         |
|                                                                                    | L_FM5  | FM5               | Head of the fifth metatarsal, dorso-lateral aspect of the fifth metatarso-phalangeal joint | X           | X         |
|                                                                                    | L_FM2  | FM2               | Base of the second metatarsal, dorso-medial aspect of the second metatarso-cuneiform joint | X           | X         |
|                                                                                    | L_PM6  | PM6               | Most distal and dorsal point of the head of the proximal phalanx of the hallux             | X           | X         |
|                                                                                    | L_FM1  | FM1               | Head of the first metatarsal, dorso-medial aspect of the first metatarso-phalangeal joint  | X           | X         |
|                                                                                    | L_P1MT |                   | Base of the first metatarsal, dorso-medial aspect of the first metatarso-cuneiform joint   | X           | X         |
|                                                                                    | L_P2MT |                   | Base of the second metatarsal, dorso-medial aspect of the second metatarso-cuneiform joint | X           | X         |
|                                                                                    | L_FNT  | FNT               | Most medial apex of the tuberosity of the navicular                                        | X           | X         |
|                                                                                    | L_FST  | FST               | Most medial apex of the sustentaculum tali                                                 | X           | X         |
|                                                                                    | R_FCC2 |                   | Proximal end of posterior aspect of calcaneus                                              | X           | X         |
|                                                                                    | R_FCC  | FCC               | Most distal point of the attachment area of the Achilles tendon on the calcaneus           | X           | X         |
|                                                                                    | R_FPT  |                   | Lateral apex of the peroneal tubercle                                                      | X           | X         |
|                                                                                    | R_FMT  | FMT               | Base of the fifth metatarsal, dorso-lateral aspect of the fifth metatarso-cuboid joint     | X           | X         |
|                                                                                    | R_FM5  | FM5               | Head of the fifth metatarsal, dorso-lateral aspect of the fifth metatarso-phalangeal joint | X           | X         |
|                                                                                    | R_FM2  | FM2               | Base of the second metatarsal, dorso-medial aspect of the second metatarso-cuneiform joint | X           | X         |
|                                                                                    | R_PM6  | PM6               | Most distal and dorsal point of the head of the proximal phalanx of the hallux             | X           | X         |
|                                                                                    | R_FM1  | FM1               | Head of the first metatarsal, dorso-medial aspect of the first metatarso-phalangeal joint  | X           | X         |
|                                                                                    | R_P1MT |                   | Base of the first metatarsal, dorso-medial aspect of the first metatarso-cuneiform joint   | X           | X         |
|                                                                                    | R_P2MT |                   | Base of the second metatarsal, dorso-medial aspect of the second metatarso-cuneiform joint | X           | X         |
|                                                                                    | R_FNT  | FNT               | Most medial apex of the tuberosity of the navicular                                        | X           | X         |
|                                                                                    | R_FST  | FST               | Most medial apex of the sustentaculum tali                                                 | X           | X         |

<sup>1</sup> Leardini, A., Sawacha, Z., Paolini, G., Ingrosso, S., Nativo, R., & Benedetti, M.G. (2007). A new anatomically based protocol for gait analysis in children. *Gait & Posture*, 26, 560-571. doi: 10.1016/j.gaitpost.2006.12.018  
[http://www.c-motion.com/v3dwiki/index.php?title=Tutorial:\\_Building\\_the\\_IOR\\_Gait\\_Model](http://www.c-motion.com/v3dwiki/index.php?title=Tutorial:_Building_the_IOR_Gait_Model)

<sup>2</sup> Sint Jan, S. Van (2007). *Color Atlas of Skeletal Landmark Definitions. Guidelines for Reproducible Manual and Virtual Palpations*. Edinburgh: Churchill Livingstone.

<sup>3</sup> Nicola Portinaro, Alberto Leardini, Artemisia Panou, Valerio Monzani and Paolo Caravaggi. Modifying the Rizzoli foot model to improve the diagnosis of pes-planus: application to kinematics of feet in teenagers. *Journal of Foot and Ankle Research* 2014, 7:57.